

BharatNet is expected to provide a minimum 2Mbps bandwidth to all rural households on-demand, and on-demand capacity to all institutions on a non-discriminatory basis. It is expected to do this by 2017. Recommendations have been submitted to the government of India and approval is awaited.

To provide online government services, National e-Governance Plan was formulated by the government's Department of Electronics and Information Technology (DeitY). As of March 2014, 133,847 common services centres (CSCs)—a core infrastructure component—have been set up at *gram panchayats*. Scope of CSCs is being broadened to provide services such as e-education and telemedicine by a village-level entrepreneur.

The Indian government introduced Digital India programme in 2014, to be managed by DeitY. Key focus areas of the programme are:

1. Digital infrastructure to all citizens, including access to high-speed broadband, cradle-to-grave digital identity and bank account

2. Governance and services on demand

3. Digital empowerment of citizens, the first initiative of which is visible on <http://mygov.nic.in>

In August 2014, the Indian government launched a financial inclusion scheme, *Jan-Dhan Yojana* (www.pmfjd.gov.in), wherein banks have been directed to open 75,000,000 zero-balance bank accounts along with a debit card for Internet banking by January 2015 in rural and underserved areas. This target has been far exceeded with around 215,100,000 accounts being opened along with a ₹ 366,000 million as total account balance.

To minimise misuse, all government schemes are linked to biometric based 12-digit unique identification numbers called *Aadhaar* provided to each citizen including infants. The unique identity crea-

tion and assignment is managed by Unique Identification Authority of India (<http://uidai.gov.in>). A little over 1000 million IDs have been issued as of April 2016 covering practically every citizen of the country.

The above initiatives coupled with the rural connectivity scenario indicate that, if rural initiatives have to have their desired impact, then broadband connectivity affordable to rural citizens and economically viable to service providers is a must.

How India is utilising the TV band spectrum

In India, frequency bands are allotted to various wireless services as per National Frequency Allocation Plan (NFAP) 2011, which also provides for future applications such as digital terrestrial broadcasting services, for example, digital TVs, mobile TVs and the like.

The only national terrestrial TV broadcasting service in India, known as *Doordarshan*, is provided by the government. Only two of the five national programmes of *Doordarshan*, namely, DD National and DD News, are broadcast throughout India using two channels called DD-I and DD-II via 1416 high-power transmitters.

DD-I is also used to broadcast regional programmes on a time-sharing basis. As each channel is of 8MHz, only 16MHz bandwidth would be in use at any given time and location.

There is a plan to switch from analogue to digital terrestrial TV transmission, and band allocation to digital broadcasting services including mobile TVs is 585MHz - 698MHz as per NFAP 2011. Studies undertaken to experimentally determine and quantify the availability of TVWS across the country, taking into consideration the pollution and protection viewpoints, show that TV band spectrum is severely underutilised with an average TVWS avail-

MECO®

SINCE 1962

LOW COST LCD / LED MODULES



LC035



LC135

Specifications

3½ Digits, 1999 Counts (max.) LCD	3½ Digits, 1999 Counts (max.) LED
LC035	LC135
0 - 200mV DC	0 - 200mV DC
0 - 2, 20, 200V DC	0 - 200µA DC
0 - 200µA DC	0 - 2, 20, 200mA DC
Auxiliary Power Supply	
9VDC ± 10%	5VDC ± 10%

Applications

- Ammeter
- Voltmeter
- Thermometer
- Power Supply
- PH Meter
- DB Meter
- Watt Meter
- LUX Meter
- LCR Meter
- Capacitance Meter
- Other Industrial and Domestic uses.

An ISO 9001:2008 Certified Company

MECO INSTRUMENTS PRIVATE LTD.

Plot No. EL-1, MIDC Electronic Zone,
TTC Industrial Area, Mahape,
Navi Mumbai - 400710 (INDIA)

Tel : 0091-22-2767 3300 (Board), 2767 3311-16 (Sales)

Fax : 0091-22-27673310, 27673330

Email : sales@meconinst.com

Web : www.meconinst.com